

Multi-Sensor Data-Driven Remaining Useful Life Prediction of Semi-Observable Systems

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Abstract—Semi-observable systems are referred to as a kind of widely used industrial equipment whose physical degradation state is only observable via shutdown inspection. To monitor the degradation process of semi-observable systems online, different types of sensors are generally employed to collect monitoring signals. Lots of studies have been conducted to fuse multi-sensor signals to predict remaining useful life (RUL). Majority of them, however, ignored the partially available state observations which can be viewed as ground truth measurements of physical degradation. To deal with this problem, this article proposes a multi-sensor data-driven RUL prediction method for semi-observable systems, which leverages degradation information from online multi-sensor signals as well as offline state observations. This method is developed based on a generalizable state-space model combined with particle filtering framework. In the framework, a state transition function is used to describe the degradation process of system states. A multidimensional measurement function is constructed to describe the mapping between states and multi-sensor signals. To enhance the performance of prediction, an algorithm named prioritized sensor group selection is also proposed to select the optimal sensor group for RUL prediction. The effectiveness of the proposed method is demonstrated using an experiment of cutting tool wear.

Index Terms—Multi-sensor data, particle filtering (PF), prognostic degradation modeling, remaining useful life (RUL) prediction, state-space model.

I. INTRODUCTION

THE performance and degradation of many industrial equipment can be tracked to some extent using sensor technology. However, the physical degradation of some equipment is generally unobservable or difficult to observe online. It can only be assessed during offline inspection. This kind of systems fall under what we refer to as semi-observable systems. More specifically, semi-observable systems are defined as a kind of equipment whose physical degradation state is unable to be inspected when the system is running. It is only observable during the shutdown period. This is a common scenario in industrial fields. Most rotating machinery can be classified into this category. For example, cracks in blades of gas turbines can only be checked by stopping the rotating system [1]. Cutting tool wear can only be measured during outages of the machine tool [2], [3].

To monitor the degradation state of semi-observable systems online, different types of sensors are generally employed to collect monitoring signals. Prognostic degradation modeling utilizes patterns and trends in the sensor signals to predict remaining useful life (RUL), which are later used to guide maintenance decisions [4], [5]. Literature in prognostic degradation modeling has focused on extracting degradation information from sensor signals and using this information to estimate system states and predict RUL [6]–[9]. However, most did not leverage valuable information from partially available state observations, which arguably can be viewed as ground truth measurements of physical degradation. Motivated by this concept, this article proposes a multi-sensor data-driven RUL prediction methodology for semi-observable systems by utilizing multi-sensor signals as well as partially available state observations. This article extends the current literature along several dimensions. First, this article provides a generalizable modeling framework for multi-sensor degradation signals and inherent system states. Second, degradation information from online monitoring signals as well as offline state inspections are utilized to predict the RUL.

Multi-sensor fusion prognostic methods can be classified into two prominent categories. The first category attempts to fuse multi-sensor raw signals or features extracted from signals into

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an aggregated (health indicator) HI, which is then used as a single input to a prognostic model. For example, Baraldi *et al.* [10] selected optimal feature subsets based on their performance metrics, and fused selected features into a HI using autoassociative kernel regression. Yu *et al.* [11] converted multi-sensor signals into an aggregated HI based on an autoencoder recurrent neural network (RNN). Tse and Wang [12] fused nine statistical features extracted from vibration signals of a slurry pump into a HI using principal component analysis (PCA). Lei *et al.* [13] assigned different weights to multiple features and fused them into a HI using a self-organizing map neural network. Guo *et al.* [14] extracted related-similarity features from signals and fused them using a RNN network for RUL prediction. Qin *et al.* [15] fused multidimensional features into a HI using the isometric mapping algorithm, and predicted RUL by a new type of long short-term memory (LSTM) network with macroscopic–microscopic attention mechanism. The basic idea of this kind of methods is to transform multi-sensor prognostics into a single degradation-based signal. This has been relatively effective in predicting RUL because it tends to fuse degradation information from different sensor resources. However, the fused HI is a virtual feature with no explicit physical meaning. Thus, it can be argued that the HI is not necessarily a precise representation of the actual degradation state.

The second category revolves around constructing a multi-input prognostic model to predict RUL by mapping the relationship between multi-sensor signals and RUL values. For example, Fang *et al.* [16] identified informative sensors via group penalization and predicted RUL by fusing the informative sensor signals using multivariate functional principle component analysis (MFPCA). He *et al.* [17] built a LSTM network to map the relationship between multi-sensor signals and RUL values of ion mill etching machines. Wang *et al.* [18] proposed a Bayesian network-based RUL prediction method, whereby the predictions of different sensors were coupled via directed acyclic graph and joint tree algorithms. Al-Dulaimi *et al.* [19] input multi-sensor signals into a LSTM network and a convolutional neural network simultaneously and fused the outputs of two models to predict RUL. This kind of methods aim at revealing the mapping between multi-sensor signals and RUL values with the help of multi-input regression models or artificial intelligent models. As a kind of data-driven methodology, it requires less expert knowledge. However, their performance is highly dependent on the quality of training data. In addition, model structures in these methods are generally prespecified making it suffer from the lack of generality and scalability.

All the above approaches assume that system states can be exactly characterized by a HI extracted from signals. Thus, they focus all efforts on extracting degradation information from sensor signals. This can be considered as an oversimplification because the relationship between sensor signals and system states is complicated due to complex signal transmission paths, attenuations, and other physical characteristics. Thus, treating the magnitude of the sensor signals to be equivalent to the state of the system can be at times unrealistic.

For semi-observable systems, partially available state observations during shutdown inspections contain significant

information about the system's degradation process. Unfortunately, this information is often ignored by most of the modeling approaches in the current literature. With that in mind, this article proposes a multi-sensor data-driven RUL prediction method for semi-observable systems. Different from existing literature, this article gives a clear distinction between system states and sensor signals. In addition to leveraging online monitoring data like the rest of studies, our proposed method also integrates offline measurements of the system state observed during offline inspections.

The underlying premise of our modeling approach is based on a state-space model combined with particle filtering (PF) framework. A state transition function that follows a generalized Wiener process is used to describe the degradation process of system states. A multidimensional measurement function is constructed to describe the mapping between system states and multi-sensor signals. Offline training data are used to estimate the correlations between multi-sensor signals and the state values. Online monitoring signals are used to estimate the hidden system state via the PF algorithm. The system's RUL is predicted as the lifetime distribution of the estimated state trajectory crossing a failure threshold. In addition to our modeling framework, we also propose a novel sensor selection strategy that we refer to as prioritized sensor group selection (PSGS). The sensitivity of each sensor is estimated, and sensors are prioritized according to their sensitivity. Then, sensors are combined following the prioritized order. The best sensor group that provides the most accurate prediction result is selected. The major contributions of this article are summarized as follows.

- 1) We develop a generalizable fusion-based prognostic model for multi-sensor applications that explicitly models the degradation process of system states and expresses multi-sensor signals as a function of system states. This modeling framework provides a more intuitive interpretation of the relationship between the degradation process of system states and the behavior of multi-sensor signals.
- 2) Degradation information involved in online monitoring multi-sensor signals as well as offline state inspections is integrated to predict RUL. The offline state inspections are the ground truth measurements of physical degradation. They provide more straightforward information about the physical degradation trajectory than multi-sensor signals.
- 3) A sensor selection algorithm referred to as PSGS is also developed. This algorithm first prioritizes different sensors by evaluating their correlation with system states, and then chooses the optimal sensor group that results in the most accurate RUL prediction. The advantage of sensor selection is that it eliminates redundant information as well as uninformative sensor signals, thus enhancing the performance of prediction.

The remainder of this article is organized as follows. Section II introduces the RUL prediction framework based on state-space models combined with PF. In Section III, the proposed multi-sensor data-driven RUL prediction method for semi-observable systems is introduced. In Section IV, the proposed method is evaluated using a degradation test of cutter wear of milling machining. Section V concludes this article.

II. RUL PREDICTION BASED ON STATE-SPACE MODELS COMBINED WITH PF

State-space modeling provides an explicit framework that relates the degradation state of a system with the observed sensor signals. It was first combined with PF for prognostics in [20], whereby PF is employed to estimate system states according to sensor signals. Up to now, lots of RUL prediction methods have been developed based on this initial framework [21]. Majority of them focused on single-sensor prognostic issues [22], [23]. This prognostic framework generally includes three major steps: model development; state estimation; and RUL prediction.

A. Model Development

A state-space model is generally composed of a state transition function and a measurement function

$$\begin{cases} x_k = f(x_{k-1}, \omega_{k-1}) \\ y_k = h(x_k, v_k) \end{cases} \quad (1)$$

where $f(\cdot)$ and $h(\cdot)$ are the state transition function and measurement function, respectively; x_k and y_k represent the system state and the degradation signal at time t_k , respectively; ω_{k-1} and v_k denote the state transition noise and the measurement noise, respectively.

B. State Estimation

As a filtering algorithm developed from the Bayesian theory, PF conducts the state estimation with the following three-step process [24].

1) Prediction: Predict the prior probability of the current state via the Chapman–Kolmogorov equation

$$p(x_k | \mathbf{y}_{1:k-1}) = \int p(x_k | x_{k-1}) p(x_{k-1} | \mathbf{y}_{1:k-1}) dx_{k-1} \quad (2)$$

where $p(x_k | x_{k-1})$ can be acquired from the state transition function in (1), and $\mathbf{y}_{1:k-1}$ denotes the historical monitoring signals up to t_{k-1} .

2) Update: The PF algorithm approximates the state probability using particles $\{\mathbf{x}_{0:k}^i\}_{i=1:N_s}$ drawn from an importance distribution $q(\mathbf{x}_{0:k} | \mathbf{y}_{1:k})$, and corresponding particle weights $\{w_k^i\}_{i=1:N_s}$ [20]

$$p(x_k | \mathbf{y}_{1:k}) \cong \sum_{i=1}^{N_s} w_k^i \delta(\mathbf{x}_{0:k} - \mathbf{x}_{0:k}^i). \quad (3)$$

Once a new measurement is available, the particle weights can be updated as follows:

$$w_k^i = \frac{w_{k-1}^i p(y_k | x_k^i) p(x_k^i | x_{k-1}^i)}{q(x_k^i | x_{k-1}^i, \mathbf{y}_{1:k})}. \quad (4)$$

3) Resampling: The resampling step is used to overcome the particle degeneracy problem in the particle updating process, i.e., most particles have negligible weights after continuous iterations. The commonly used resampling algorithm is systematic resampling [24], which repeats particles many times if they have large weights. This algorithm is effective in dealing with particle degeneracy, but leads to another sample impoverishment problem, i.e., most samples are repeated points from the same

particle thus reducing the diversity among samples. To deal with this problem, a fuzzy resampling algorithm is proposed in [25], which increases the diversity of samples by adding random noise to repeated samples. Herein, the fuzzy resampling algorithm is employed.

C. RUL Prediction

Under the concept of first passage time, the RUL amounts to the first time it takes the system state to cross a prespecified failure threshold from its current state

$$L = \inf \{l : x(l + t_k) \geq D | x_k < D\} \quad (5)$$

where L denotes the RUL at t_k , $\inf\{\cdot\}$ denotes the inferior limit of a variable, $x(l + t_k)$ is the system state at $l + t_k$, and D is the failure threshold.

The RUL distribution is determined by the expression of state transition functions. Closed form solution of RUL distribution is only available for specific state transition functions, such as age-dependent Wiener processes [26] and geometric gamma processes [27]. Otherwise, numerical simulation is employed to estimate the RUL distribution. In [28], a generalized age- and state-dependent Wiener process model was developed, and a degradation process simulation (DPS) algorithm was proposed to approximate the RUL distribution. It is consistent with our intention to construct a generalizable modeling framework. Thus, it will be employed in this study.

III. MULTI-SENSOR DATA-DRIVEN RUL PREDICTION FOR SEMI-OBSERVABLE SYSTEMS

In this section, a multi-sensor data-driven RUL prediction method is proposed for semi-observable systems based on the state-space model combined with PF framework. Fig. 1 outlines the flowchart of the proposed method. At first, a generalizable multi-dimensional state-space model is developed to describe the degradation process of the system state and its relationship with multi-sensor signals. After that, model parameters are estimated according to offline training data. Then, the PSGS algorithm is proposed to select informative sensors. Finally, the system state and degradation rate are estimated jointly using a PF algorithm [25], and the RUL is predicted. The details of each step are introduced as follows.

A. Model Development

The degradation process of the system state is described using a generalized Wiener process model [25], [28]

$$x(t) = x(0) + u \int_0^t \omega(x(\tau), \tau, \boldsymbol{\theta}) d\tau + \sigma_B B(t) \quad (6)$$

where $x(t)$ is the system state at time t , $\omega(x(\tau), \tau, \boldsymbol{\theta})$ is an age- and state-dependent function denoting the degradation rate, u is a parameter describing the unit-to-unit variability, $\boldsymbol{\theta}$ is a parameter vector, $B(t)$ is a standard Browning motion (BM), and σ_B is the standard deviation of the diffusion term.

According to the derivation result in [25] and [28], the RUL of the degradation process in (6) has the following probability

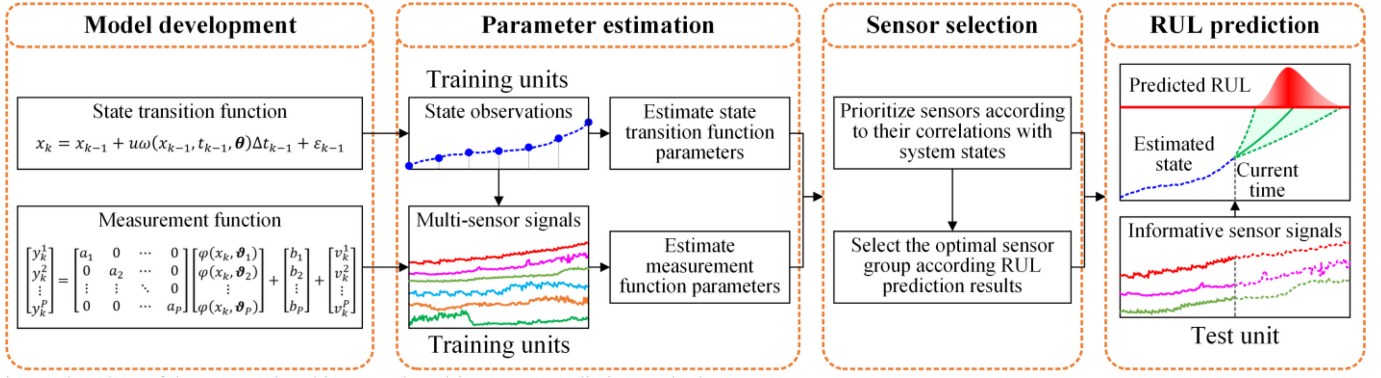


Fig. 1. Flowchart of the proposed multi-sensor data-driven RUL prediction method.

density function (PDF): Eq. (7) shown at the bottom of this page.

The above analytical solution is unavailable in real applications, since it requires the future degradation trajectory $\{x(\tau + t_k)\}_{0 \leq \tau \leq l}$ which is unknown in practice. The RUL PDF, however, can be approximated numerically by the DPS algorithm [28]. The major idea of this algorithm is to generate a series of possible future degradation trajectories through Monte Carlo simulation and approximate the distribution of the RUL by inputting the simulated degradation trajectories into the analytical solution.

Without loss of generality, the sensor number is represented as a variable P . The measurement values of P sensors can be expressed using a multidimensional measurement function depending on the system state

$$\begin{bmatrix} y_k^1 \\ y_k^2 \\ \vdots \\ y_k^P \end{bmatrix} = \begin{bmatrix} a_1 & 0 & \cdots & 0 \\ 0 & a_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_P \end{bmatrix} \begin{bmatrix} \varphi(x_k, \vartheta_1) \\ \varphi(x_k, \vartheta_2) \\ \vdots \\ \varphi(x_k, \vartheta_P) \end{bmatrix} + \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_P \end{bmatrix} + \begin{bmatrix} v_k^1 \\ v_k^2 \\ \vdots \\ v_k^P \end{bmatrix} \quad (8)$$

where $\varphi(\cdot)$ is a nonlinear function of states, ϑ_p denotes the parameter vector corresponding to sensor p , a_p and b_p are the scale and location parameters of sensor p , respectively. Since different sensor signals are various in transmission paths, fault sensitivities, etc., their signal curves generally present different degradation behaviors. The variation of degradation behaviors can be formulated by assigning different parameters a_p , b_p , and ϑ_p into different sensors. In addition, since multi-sensor signals are responses of the same degradation process, they may suffer similar interference from the outside environment. Therefore, they also present a high correlation with each other, especially in their measurement noises. To describe the correlation between

different sensors, the noise vector $V_k = (v_k^1, v_k^2, \dots, v_k^P)'$ is formulated as a P -dimensional multivariate normal distribution $N(0, \Sigma)$ with

$$\Sigma = \begin{bmatrix} \sigma_1^2 & \sigma_{2,1} & \cdots & \sigma_{P,1} \\ \sigma_{1,2} & \sigma_2^2 & \cdots & \sigma_{P,2} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1,P} & \sigma_{2,P} & \cdots & \sigma_P^2 \end{bmatrix} \quad (9)$$

where σ_p^2 ($p = 1, 2, \dots, P$) is the variance of the measurement noise of sensor p , $\sigma_{i,j}$ ($i \neq j$) is the covariance of the measurement noises between sensor i and j .

Let $\Phi_k = (\varphi(x_k, \vartheta_1), \varphi(x_k, \vartheta_2), \dots, \varphi(x_k, \vartheta_P))'$, $A = \text{diag}(a_1, a_2, \dots, a_P)$ which denotes a positive diagonal matrix with elements $a_1, a_2, \dots, a_P$, $B = (b_1, b_2, \dots, b_P)'$. The following multidimensional state-space model is developed

$$\begin{cases} x_k = x_{k-1} + u\omega(x_{k-1}, t_{k-1}, \theta)\Delta t_{k-1} + \varepsilon_{k-1} \\ Y_k = A\Phi_k + B + V_k \end{cases} \quad (10)$$

where the state transition function is a discrete difference expression of (6), $\Delta t_k = t_k - t_{k-1}$ is the time interval, ε_{k-1} is a random sample from $N(0, \sigma_B^2 \Delta t_k)$, and the measurement function is simplified from (8).

B. Parameter Estimation

Assume that we have captured system state observations and degradation signals of N training units. For unit n , the system states $(x_{n,1}, x_{n,2}, \dots, x_{n,K_n})'$ are observed at shutdown inspection time sequence $(t_{n,1}, t_{n,2}, \dots, t_{n,K_n})'$, where $x_{n,k}$ denotes the state observation at $t_{n,k}$, K_n denotes the total number of samples. The samples of sensor p at the same inspection time is indicated as $(y_{n,1}^p, y_{n,2}^p, \dots, y_{n,K_n}^p)'$.

$$f(l|x_k) \cong \left(\frac{D - x_k - u \int_0^l \omega(x(\tau + t_k), \tau + t_k, \theta) d\tau}{l} + u\omega(x(l + t_k), l + t_k, \theta) \right) \times \frac{1}{\sqrt{2\pi\sigma_B^2 l}} \exp \left(- \frac{(D - x_k - u \int_0^l \omega(x(\tau + t_k), \tau + t_k, \theta) d\tau)^2}{2\sigma_B^2 l} \right) \quad (7)$$

1) Estimation of State Transition Function Parameters:

Parameter u describes the unit-to-unit variability of degradation rates, thus it has distinct values among different units, which can be denoted as $u_1, u_2, \dots, u_N$. The unknown model parameters in the state transition function include $u_1, u_2, \dots, u_N, \theta$ and σ_B^2 . These parameters can be estimated using the unit maximum likelihood estimation algorithm [25]. Let $\Delta \mathbf{X}_n = (x_{n,2} - x_{n,1}, x_{n,3} - x_{n,2}, \dots, x_{n,K_n} - x_{n,K_n-1})'$. According to the independent characteristic of the BM process, $\Delta \mathbf{X}_n$ follows the multivariate normal distribution $N(u_n \boldsymbol{\omega}_n(\theta), \sigma_B^2 \boldsymbol{\Omega}_n)$, where $\boldsymbol{\omega}_n(\theta) = (\omega_{n,1}(\theta) \Delta t_{n,1}, \omega_{n,2}(\theta) \Delta t_{n,2}, \dots, \omega_{n,K_n}(\theta) \Delta t_{n,K_n})'$.

$$\boldsymbol{\Omega}_n = \begin{bmatrix} \Delta t_{n,1} & 0 & \cdots & 0 \\ 0 & \Delta t_{n,2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \Delta t_{n,K_n-1} \end{bmatrix}. \quad (11)$$

The log-likelihood function of the unknown parameters $\Theta = (u_1, u_2, \dots, u_N, \theta, \sigma_B^2)$ is formulated as

$$\begin{aligned} \mathcal{L}(\Theta | \mathbf{X}) = & -\frac{\ln(2\pi) + \ln \sigma_B^2}{2} \sum_{n=1}^N (K_n - 1) - \frac{1}{2} \sum_{n=1}^N \ln |\boldsymbol{\Omega}_n| \\ & - \frac{\sum_{n=1}^N (\Delta \mathbf{X}_n - u_n \boldsymbol{\omega}_n(\theta))' \boldsymbol{\Omega}_n^{-1} (\Delta \mathbf{X}_n - u_n \boldsymbol{\omega}_n(\theta))}{2\sigma_B^2}. \end{aligned} \quad (12)$$

Calculate the first partial derivatives of (12) with respect to $u_1, u_2, \dots, u_N$, and σ_B^2 , and let them equal zeros. The estimations of $u_1, u_2, \dots, u_N$, and σ_B^2 are expressed as functions of θ :

$$u_n(\theta) = \frac{\boldsymbol{\omega}_n'(\theta) \boldsymbol{\Omega}_n^{-1} \Delta \mathbf{X}_n}{\boldsymbol{\omega}_n'(\theta) \boldsymbol{\Omega}_n^{-1} \boldsymbol{\omega}_n(\theta)}, \quad n = 1, 2, \dots, N \quad (13)$$

$$\begin{aligned} \sigma_B^2(\theta) = & \frac{\sum_{n=1}^N (\Delta \mathbf{X}_n - u_n \boldsymbol{\omega}_n(\theta))' \boldsymbol{\Omega}_n^{-1} (\Delta \mathbf{X}_n - u_n \boldsymbol{\omega}_n(\theta))}{\sum_{n=1}^N (K_n - 1)}. \end{aligned} \quad (14)$$

Substitute (13) and (14) into (12). The estimation result of θ is solved by the following criterion:

$$\begin{aligned} \max_{\theta} \left\{ & -\frac{(\ln(2\pi) + \ln \sigma_B^2(\theta) + 1) \sum_{n=1}^N (K_n - 1)}{2} \right. \\ & \left. - \frac{\sum_{n=1}^N \ln |\boldsymbol{\Omega}_n|}{2} \right\}. \end{aligned} \quad (15)$$

Then, the estimation results of $u_1, u_2, \dots, u_N$, and σ_B^2 are calculated by submitting $\hat{\theta}$ into (13) and (14), respectively.

2) Estimation of Measurement Function Parameters:

According to the definition of the measurement function (8), the unique degradation behavior of each sensor is determined by its own parameters a_p, b_p , and ϑ_p . The correlation between different sensors is quantified by the covariance matrix $\boldsymbol{\Sigma}$. To be consistent with the variation and correlation between different sensors, we estimate the model parameters of the measurement function following two steps. In the first step, each sensor signal

is analyzed separately to estimate the model parameters a_p, b_p , and ϑ_p . In the second step, all sensor signals are analyzed jointly to estimate the covariance matrix $\boldsymbol{\Sigma}$.

Let us first focus on an individual sensor signal. According to (8), each sensor satisfies the following relationship:

$$y_{n,k}^p = a_p \varphi(x_{n,k}, \vartheta_p) + b_p + v_{n,k}^p. \quad (16)$$

The model parameters a_p, b_p and ϑ_p can be estimated with the following optimization criterion:

$$\min_{a_p, b_p, \vartheta_p} \left\{ \sum_{n=1}^N \sum_{k=1}^{K_n} (y_{n,k}^p - a_p \varphi(x_{n,k}, \vartheta_p) - b_p)^2 \right\}. \quad (17)$$

The multi-sensor measurement vector of unit n at $t_{n,k}$, i.e., $\mathbf{Y}_{n,k} = (y_{n,k}^1, y_{n,k}^2, \dots, y_{n,k}^P)'$, follows the multivariate normal distribution $N(\mathbf{A}\boldsymbol{\Phi}_{n,k} + \mathbf{B}, \boldsymbol{\Sigma})$. The log-likelihood function of $\boldsymbol{\Sigma}$ is expressed as

$$\begin{aligned} \mathcal{L}(\boldsymbol{\Sigma} | \mathbf{Y}) = & \sum_{n=1}^N \left(-\frac{PK_n}{2} - \frac{K_n}{2} \ln |\boldsymbol{\Sigma}| \right. \\ & \left. - \frac{\sum_{k=1}^{K_n} (\mathbf{Y}_{n,k} - \mathbf{A}\boldsymbol{\Phi}_{n,k} - \mathbf{B})' \boldsymbol{\Sigma}^{-1} (\mathbf{Y}_{n,k} - \mathbf{A}\boldsymbol{\Phi}_{n,k} - \mathbf{B})}{2} \right). \end{aligned} \quad (18)$$

The maximum likelihood estimation of $\boldsymbol{\Sigma}$ is the value that satisfies $\partial \mathcal{L}(\boldsymbol{\Sigma} | \mathbf{Y}) / \partial \boldsymbol{\Sigma} = 0$

$$\begin{aligned} \hat{\boldsymbol{\Sigma}} = & \frac{\sum_{n=1}^N \sum_{k=1}^{K_n} (\mathbf{Y}_{n,k} - \mathbf{A}\boldsymbol{\Phi}_{n,k} - \mathbf{B}) (\mathbf{Y}_{n,k} - \mathbf{A}\boldsymbol{\Phi}_{n,k} - \mathbf{B})'}{\sum_{n=1}^N K_n}. \end{aligned} \quad (19)$$

C. Sensor Selection

The task of sensor selection is to find informative sensors with which the methodology can get the most accurate RUL prediction. A straightforward way is to try all possible groups and select the optimal one according to their performance. In theory, P sensors totally generate $2^P - 1$ random groups (including the cases with a single sensor). The group number increases exponentially with the sensor number. It will be time consuming to try all possible groups in high-dimensional cases. To reduce computations and guarantee the validity of the selection result, we propose a PSGS algorithm. This algorithm first prioritizes P sensors according to their correlation coefficients with system states, and then generates P sensor groups following their prioritization. The sensor group that provides the most accurate RUL prediction is finally selected as the optimal one. The detailed process of the PSGS algorithm is explained as follows.

Sensor signals generally present nonlinear correlation with systems states. To evaluate the correlation between the system states and different sensor signals quantitatively, a commonly used nonlinear correlation metric, i.e., the absolute Spearman

coefficient (ASC), is calculated as follows:

$$\rho_p = \frac{1}{N} \sum_{n=1}^N \frac{\left| \sum_{k=1}^{K_n} (\tilde{y}_{n,k}^p - \bar{y}_n^p) (\tilde{x}_{n,k}^p - \bar{x}_n^p) \right|}{\sqrt{\sum_{k=1}^{K_n} (\tilde{y}_{n,k}^p - \bar{y}_n^p)^2 \sum_{k=1}^{K_n} (\tilde{x}_{n,k}^p - \bar{x}_n^p)^2}} \quad (20)$$

where ρ_p indicates the ASC between the measurement sequence of sensor p and the state sequence, whose score is within interval $[0, 1]$, $\tilde{x}_{n,k}$ and $\tilde{y}_{n,k}^p$ are the ranks of $x_{n,k}$ and $y_{n,k}^p$ in their ascending sequences, $\bar{x}_n^p$ and $\bar{y}_n^p$ are the mean of $\tilde{x}_{n,k}$ and $\tilde{y}_{n,k}^p$, respectively. If the ranks of measurements $\tilde{x}_{n,k}$ and $\tilde{y}_{n,k}^p$ are changed to the raw signals $x_{n,k}$ and $y_{n,k}^p$, (20) will become the Pearson coefficient. The advantage of the Spearman coefficient compared with the Pearson coefficient is that it is more suitable to deal with nonlinear problems [5], [13]. Thus, it is employed in this article.

A higher ASC score indicates that the sensor is more informative to the degradation of the system state. Thus, the P sensors are sorted in descending order following their ASC scores. Sensors in front of the order will be chosen preferentially to predict the RUL. Following this rule, we generate P different sensor groups by adding one sensor at each time iteratively. In particular, the p th sensor group is composed of the first p sensors in the priority order. After that, each group is used to predict the RUL of the training units. The PF with a fuzzy resampling algorithm [25] is employed to estimate the degradation rate and system state jointly. For sensor group p , the RUL prediction process is conducted as follows.

At the initial time, a series of parameter and state particles $\{u_0^i, x_0^i\}_{i=1:N_s}$ are generated, where N_s is the particle number. u_0^i and x_0^i are sampled using a linear interpolation method [25] from the sequences $\{\hat{u}_n\}_{n=1:N}$ and $\{\hat{x}_{n,0}\}_{n=1:N}$, respectively. Particle weights are initialized as $\{w_0^i = 1/N_s\}_{i=1:N_s}$. At each time, the state particle is one-step transmitted following the state transition function:

$$x_k^i = x_{k-1}^i + u_{k-1}^i \omega \left(x_{k-1}^i, t_{k-1}, \hat{\theta} \right) \Delta t_{k-1} + \varepsilon_{k-1}^i. \quad (21)$$

The particle weights are updated [24] according to the observations of the sensor group p using

$$w_k^i = \frac{w_{k-1}^i}{\sqrt{(2\pi)^p |\Sigma_p|}} \times \exp \left(-\frac{(Y_{p,k} - A_p \Phi_{p,k}^i - B_p)' (Y_{p,k} - A_p \Phi_{p,k}^i - B_p)}{2\Sigma_p} \right) \quad (22)$$

where $Y_{p,k}$ is the observations at t_k from group p . A_p , B_p , $\Phi_{p,k}^i$, and Σ_p are the submatrices of A , B , Φ_k^i , and Σ , respectively, formed by the elements corresponding to the sensors in group p .

The particles are resampled using the fuzzy resampling algorithm [25] and the particle weights are reset to be $w_k^i = 1/N_s$. The resampled particles are denoted as $\{u_k^i, x_k^i\}_{i=1:N_s}$. The median of the particles $\hat{u}_k$ and $\hat{x}_k$ are employed as the estimation results of the parameter and the system state, respectively. The

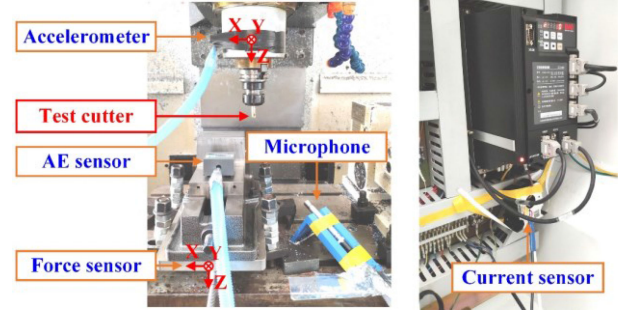


Fig. 2. Experimental platform and the location of five sensors.

RUL PDF is predicted by the DPS algorithm based on the updated parameter and state.

After RUL prediction, the performance of different sensor groups is evaluated by calculating the weighted absolute relative error (WARE) [5] between the expectation of the prediction result and the actual failure time

$$\text{WARE}_n = \frac{1}{K_n} \sum_{k=1}^{K_n} \frac{t_{n,k} |T_n - \hat{l}_{n,k} - t_{n,k}|}{T_n^2} \times 100\% \quad (23)$$

where T_n is the actual failure time of unit n , and $\hat{l}_{n,k}$ is the expectation of the predicted RUL of unit n at $t_{n,k}$.

A lower WARE score corresponds to a more accurate RUL prediction result. The mean of the WARE scores of N training units are calculated. The sensor group that occupies the lowest mean WARE score is selected as the optimal one, which is employed to predict the RUL of test units in the following process.

D. RUL Prediction

The RUL of a test unit is predicted in real time by fusing its online monitoring signals of the selected sensor group. We denote the selected sensor group as S . The RUL prediction process using sensor group S is the same as that using group p in the previous subsection. We only need to input the signals and the parameter vectors of the selected sensor group into the particle weight updating (22). In particular, the observed sensor signals are changed as $Y_{S,k}$, and the parameters in the multidimensional measurement function become A_S , B_S , $\Phi_{S,k}$, and Σ_S , respectively. After that, parameter and state particles are updated using the fuzzy resampling algorithm. The RUL PDF is predicted using the DPS algorithm based on the updated particles.

IV. EXPERIMENTAL DEMONSTRATION

A. Experimental Data Introduction

The experimental data were captured from a milling machine as shown in Fig. 2. The basic setup of the platform is a VMC850 computer numerical control machine. The tool wear during the milling process was monitored by five different types of sensors. A PCB 356A15 three-axis accelerometer was mounted on the spindle. A KISTLER 9257B three-axis force sensor was

TABLE I
PARAMETER SETTING OF THE MILLING CUTTER WEAR EXPERIMENT

Parameter	Condition 1	Condition 2
Rotating speed (rpm)	4500	3500
Cutting speed (mm/min)	400	300
Depth of cut (mm)	1.5	1.75
Material of workpiece	Steel 45	Steel 45
Size of workpiece (mm ³)	100×60×60	100×60×60

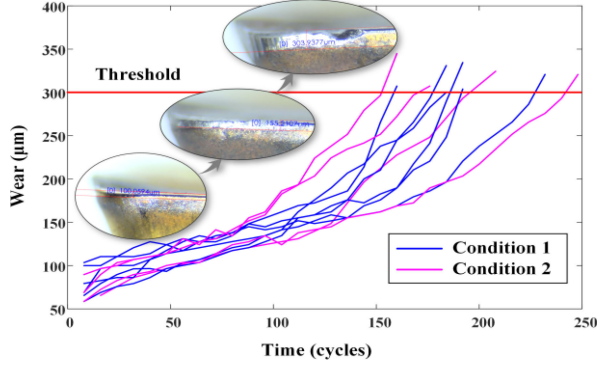


Fig. 3. Flank wear growth curves of ten cutters.

installed underneath the workpiece. Both the accelerometer and the force sensor have three different direction channels designated X, Y, and Z. An Anyway CSA201-P030T01 current sensor was used to monitor the current of the spindle motor. A Brüel and Kjær 4966 microphone and a PAC WD acoustic emission (AE) sensor were mounted beside the workpiece to capture the sound signal and AE signal, respectively. The five sensors totally generated nine signal channels designated P1~P9. Sensor signals were sampled at each cutting cycle. The AE sensor sampled with 1 MHz sampling frequency and 1 s time length. The remaining sensors sampled with 10 kHz frequency and more than 5 s time length.

Two different working conditions were set based on industrial recommendation. The parameter values under two conditions are given in Table I. According to the tool-life criterion in ISO 8688-1, the experiment was stopped once the width of the flank wear exceeded 300 μm. There are six failed cutters under condition 1 designated 1-1~1-6, and four failed cutters under condition 2 designated 2-1~2-4. The flank wear of each cutter was measured between every eight cycles except 2-1 which is measured between every 16 cycles. The flank wear growth curves of ten cutters were recorded during their whole lifetimes, as shown in Fig. 3. Snapshots of a cutter at three different degradation stages are presented in Fig. 3, as well. The root mean square (rms) of each signal is calculated and displayed in Fig. 4.

B. Informative Sensor Selection for Cutter Wear Degradation

To describe the time-varying degradation rate of the cutters, the degradation rate $\omega(x(\tau), \tau, \theta)$ in (6) is formulated using three different expressions [28], i.e., $\theta\tau^{\theta-1}$, $\theta\exp(\theta\tau)$ and $x(\tau)^{\theta}$.

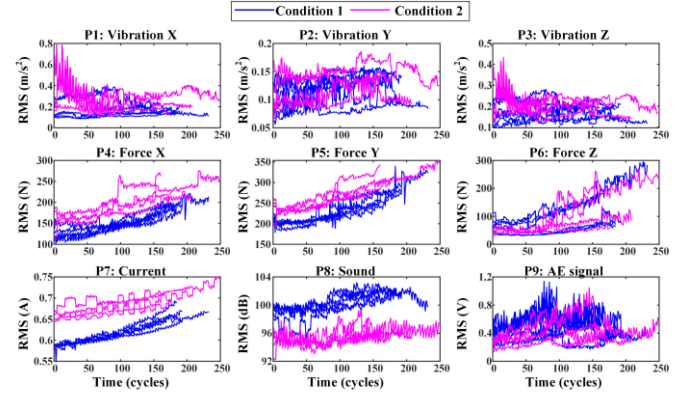


Fig. 4. RMS values of nine sensor signals.

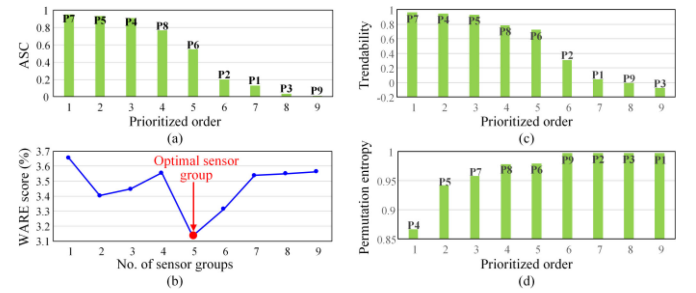


Fig. 5. Sensor selection process in the first round under condition 1. (a) ASC scores. (b) WARE scores. (c) Trendability scores. (d) Permutation entropy scores.

Among them, the exponent-type function $\theta\exp(\theta\tau)$ acquires the lowest Akaike information criterion (AIC) score, which means that it is best fitted with the degradation curves. Therefore, it is employed in the state transition function. The relationship between the state and the measurement is expressed using a polynomial function $\varphi(x_k, \vartheta) = x_k^{\vartheta}$. It is seen from Fig. 4 that the working condition has obvious impact on the behavior of sensor signals. Therefore, the experimental data under two different conditions are analyzed separately using leave-one-out cross-validation. In particular, one cutter is used as the test unit in each round. Meanwhile, the remaining cutters are used as the training set. Take the first round under condition 1 for example, whereby the degradation data of 1-2~1-6 are used to estimate model parameters and select informative sensors, and the RUL of 1-1 is predicted.

To provide the details of the sensor selection process in the first round under condition 1, the ASC scores of the nine sensors and WARE scores of the nine groups are shown in Fig. 5 (a) and (b), respectively. It is seen from Fig. 5 (a) that, the five sensors P4~P8 acquire higher ASC scores than other ones. The result is consistent with the rms curves in Fig. 4, where the five sensors show more obvious increasing trends than others. To further validate the correctness of the prioritized order, we compare the result with other two feature optimization metrics, i.e., the trendability [29] and improved permutation entropy [30]. Trendability evaluates the correlation between the feature sequence and time. A higher score corresponds to a better feature. The improved

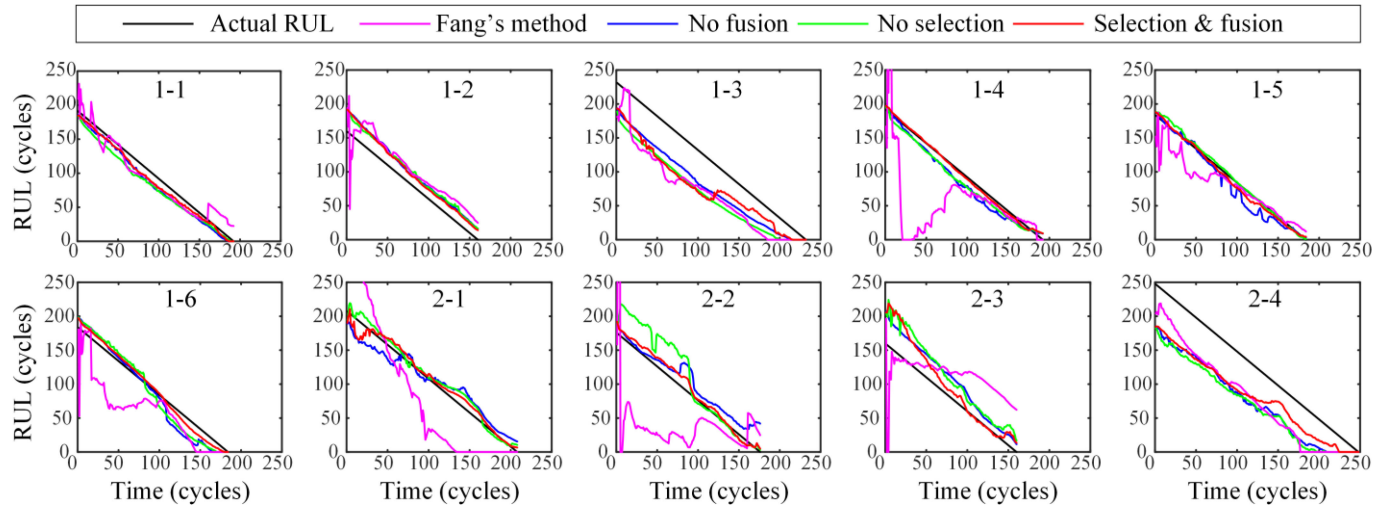


Fig. 6. RUL prediction results of ten cutters.

permutation entropy reflects the monotony of the feature series, where a smaller value is expected. Their prioritized order results are given in Fig. 5(c) and (d), respectively. Although these two metrics present different orders with our method, they rank the five sensors P4~P8 in front of the order. It indicates that these two metrics can also identify sensitive sensors effectively. However, they cannot determine an exact number of sensors. In Fig. 5(b), our method acquires the lowest WARE score in the fifth group, which means that the combination of the five sensors P4~P8 acquires the most accurate RUL prediction result. Therefore, this informative sensor group is selected in the first-round validation. The same sensor group is also selected in the remaining five rounds.

C. RUL Prediction of Cutters

After sensor selection, the RUL of the test cutter is predicted using the selected informative sensors. For comparison, we predict the RUL using other three methods as well. Two of them use the same state-space model-based algorithm, but different sensor groups with our method. In particular, one method designated “no fusion” only uses the first sensor that performs best in Fig. 5(a). The other method designated “no selection” uses all nine sensors to predict RUL. Our method is labeled “selection and fusion.” We also choose a benchmark method proposed by Fang *et al.* [16], which fuses multi-sensor signals using MFPCA. The reason we choose this method is that it has similar flowchart and working process with our method. It does not use any ground truth physical observations. Thus, this method is suitable to validate the contribution of our method in leveraging state observations and sensor signals.

The RUL prediction results of the ten cutters are shown in Fig. 6. Table II gives the computation times of the cross-validation in a platform with Intel I7-7700HQ CPU and 8G RAM. Under each condition, the method takes less time than Fang’s method and the “no selection” method. The “no fusion” method takes the least time. Because it only uses

TABLE II
COMPARISON OF THE COMPUTATION TIME UNDER CONDITION 1

No. of conditions	Fang’s method	No fusion	No selection	Selection & fusion
Condition 1	2.54 s	1.94 s	2.45 s	2.27 s
Condition 2	1.74 s	1.46 s	1.78 s	1.71 s

TABLE III
COMPARISON OF WARE SCORES OF THE RUL PREDICTION RESULTS

No. of cutters	Fang’s method	No fusion	No selection	Selection & fusion
1-1	4.00	3.21	3.01	1.95
1-2	8.16	5.89	5.66	4.89
1-3	8.90	8.83	6.97	6.72
1-4	6.89	2.73	3.02	0.97
1-5	2.67	0.61	2.69	1.21
1-6	7.09	4.27	4.10	2.01
2-1	10.18	2.01	3.18	1.45
2-2	11.82	4.12	5.25	1.94
2-3	16.03	7.55	6.58	3.45
2-4	8.69	9.26	8.68	6.82
Mean	8.44	4.85	4.91	3.14
Variance	14.41	8.64	4.19	4.99

one-dimensional sensor signals thus taking less resources than high-dimensional computation. It should be mentioned that the computation time involves both model training and testing processes. In real applications, only the testing process needs to be conducted online. The computation time of each prediction step is at millisecond level. Therefore, the proposed method is able to provide real-time prediction results in practice.

To compare the performance of the four methods quantitatively, the WARE scores of the predicted RUL is calculated using (23). The results are given in Table III. It is seen that the proposed method acquires the lowest WARE score for nine out of ten cutters. Its mean WARE value is also smaller than other three competitors. The “no selection” method achieves the lowest

variance. The results indicate that more sensors may present more stable RUL predictions. However, the method presents the most accurate RUL prediction result by fusing informative sensor signals.

The superiority of the method compared with the other three methods can be explained as follows. Fang's method only uses sensor signals for RUL prediction. It ignores the valuable information in state observations. In addition, this method is highly dependent on the quality and quantity of training datasets. The experimental data in this article only provide five and three available training units, which is not enough for this method to train a good prognostic model. The "no fusion" method predicts the RUL using an individual sensor, which may provide incomplete degradation information resulting in inaccurate prediction results. Conversely, the "no selection" method fuses all sensor signals for RUL prediction. However, noninformative sensors may reduce the performance of the method. The method selects informative sensors following their priority order and fuses them using a generalizable state-space modeling framework. It promises the validity of sensor selection and fusion. As a result, it acquires the most accurate RUL prediction result among the four competitors.

V. CONCLUSION

This article proposed a multi-sensor data-driven RUL prediction method for semi-observable systems. The main contribution of the proposed method was concluded as follows. A generalizable fusion-based prognostic model was developed, which provides a more intuitive interpretation of the relationship between the degradation process of system states and the behavior of multi-sensor signals. Degradation information from online monitoring multi-sensor signals as well as offline state observations are integrated to predict RUL. An experimental dataset of cutter wear was used to validate the effectiveness of the proposed method. The result indicates that the method selects informative sensors effectively from multi-sensor networks and improves the RUL prediction accuracy by fusing informative sensor signals.

In the future, we will improve the method by introducing a multistage-varying measurement function. A basic assumption in our method was that there is a consistent relationship between states and sensor signals during the whole degradation process. However, the relationship between states and signals may present varying trends under different stages. It will be necessary for these cases to detect the critical state node and correct the model adaptively. In addition, we will employ our method into other semi-observable systems, such as gearboxes and gas turbines, to validate its generality in different fields.

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